PRIMARY SCIENCE

TOPICAL REVISION

WEATHER

SUMMARIZED SCIENCE REVISION NOTES

WEATHER AND THE SKY

The space that surrounds the earth and contains air is called atmosphere.

CLOUDS

Clouds are grouped according to their appearance, shape and height.

TYPES OF CLOUDS

Cumulus

- They are thick, white, feathery clouds
- They look like huge heaps of cotton wool
- They have a flat base
- They are common in fine weather
- They are found high in the sky

Nimbus

- They are found low in the sky
- They are dark or grey in colour
- They are heavy, mountainous clouds.
- They are rain-laden clouds
- They bring heavy rains

Cirrus clouds

- They are small and appear like white separate feathers, very high up the sky

Stratus

- They appear like layers on top of each other.
- They are horizontal and are seen over high grounds

STARS

- Twinkle and produce light and heat

THE SUN

- It is a star
- It produces light and heat
- It appears bigger than other stars because it is nearer the earth
- It is the chief source of heat and light
- The study of the heavenly bodies (stars and other objects in the sky) is called Astronomy
- People who study heavenly bodies are called Astronomers

THE MOON

- It does not produce its own light but reflects from the sun.
- The different shape and size of the moon at different times of the month is called the Phase of the moon.
- The path followed by the moon as it rotates is called Orbit
- The moon takes 29 ½ days to go round the earth

Diagram showing phases of the moon.

Weather is the condition of the atmosphere at a particular time in a certain place over a short period of time.

ELEMENTS OF WEATHER/ CONDITIONS OF THE ATMOSPHERE

- Cloudy
- Sunny
- Windy
- Calm
- Humid
- Hot/cold – Temperature
- Rainy
- Snowy
- Humidity is the amount of water vapour in the atmosphere
- Study of the weather is called Meteorology.
- Experts or scientists who study the weather are called Meteorologists
- Study of the weather is done in a weather station
- Weather is measured using weather instruments.

EXAMPLES OF WEATHER INSTRUMENTS

1. Rain gauge

- Measure the amount of rainfall in millimeters (mm)
- Place on an open ground (field) away from trees and buildings
- Should be placed 30cm below the ground level in order to
    - a) Make it firm
    - b) Reduce the rate of evaporation
- The top of the rain collector should also raised above the ground level to avoid splashing of the rain drops.
- Measurements should be taken on a daily (everyday) basis at the same time.

2. **Wind vane:**

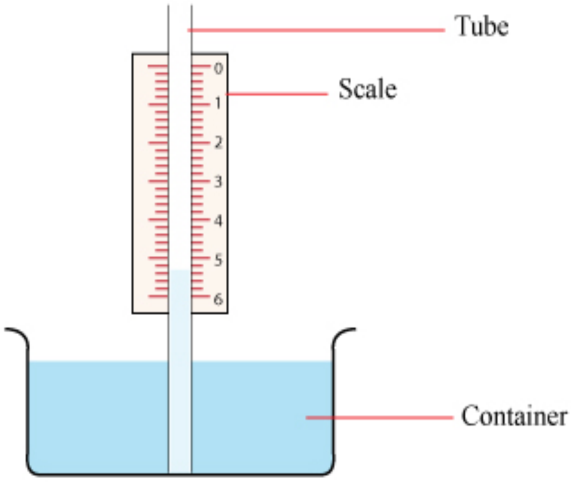
- Shows the direction of wind
- Tells the direction of wind
- Place in an open field with free movement of wind
- Arrowhead points where the wind is blowing from
- Tail points where the wind is blowing to
- Tail is bigger than the arrowhead
- The rotating horizontal arm is fixed with a biro pen cap (top) for it to rotate freely.

3. **Windsock:-** Tells two aspects of weather:
   - Strength of wind
   - Direction wind
   - Open on both sides
   - Mouth is wider than the end (foot)
   - Painted with black and white stripes to make it visible from far.
   - Used mainly in Airports/Air strips.

4. **Thermometers:-** Used to measure temperature
   - Measure temperature in degrees Celsius (0◦c)
   - Temperature is the hotness or coldness of a place or object.
   - A clinical thermometer measures body temperature
   - Weather thermometers include:
     a) Air thermometer
     b) Liquid thermometer

N.B Liquids used in thermometers (Mercury / alcohol)

Liquid Thermometer



• Works on the principle of liquids expanding when heated and contracting when cooled down.
• Increase in temperature makes the coloured water rise in the narrow tube – the scale reads an increase in temperature.
• Decrease in temperature makes the coloured water in the tube drop – The scale reads a decrease in temperature.

N.B For accurate results, the tube used in thermometers should be very narrow

• Works on the principle of gases expanding when heated and contracting when cooled down.
• The scale of air thermometer reads from top to bottom
• The increase in temperature results in a drop of coloured water in the narrow tube used – the scale shows an increase in temperature.
• The decrease in temperature makes the coloured water raise in the tube – the scale shows a decrease in temperature.

a). Mercury
b). Alcohol

Anemometer – measures speed of the wind.

Barometer – measures atmospheric air pressure

Hygrometer – measures the amount of water vapour in the atmosphere (humidity)

WEATHER AND THE SOLAR SYSTEM

Stars – bright objects seen at night that twinkle to produce their own light.

- They are not seen during the day because of the bright light from the sun.

Planets – Bright objects which do not twinkle

They give steady light

Planets do not produce their own light; they reflect from the sun.

The study of heavenly bodies e.g. stars, planets and moon is called Astronomy.

Scientists who study heavenly bodies are called Astronomers.

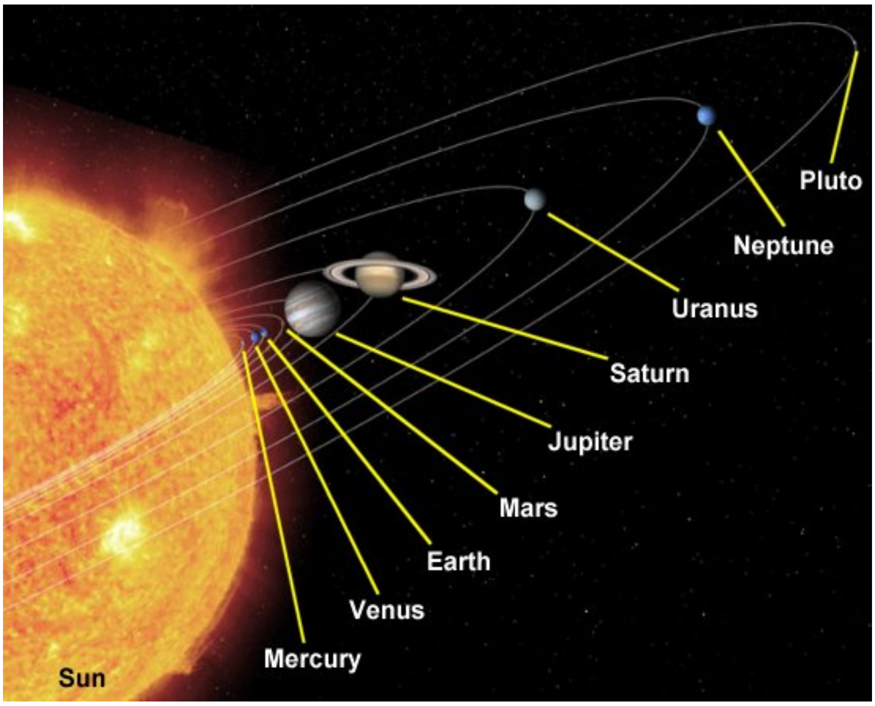
THE SOLAR SYSTEM

The sun and the planets make up the solar system.

There are eight planets which revolve round the sun starting with the one that is nearest the sun.

1. Mercury
2. Venus
3. Earth
4. Mars
5. Jupiter
6. Saturn
7. Uranus
8. Neptune

The Solar System



Mercury - the smallest planet

- Nearest to the sun
- seen in the west just as the night begins (after sunset)

Venus - seen early in the morning (before dawn) in the East

- The brightest planet

Jupiter - the largest planet

- The planet with the largest number of moons i.e. 12 moons

Saturn - Has a large major ring or halo around it.

N.B Apart from Mercury and Venus all other planets have moons.

STEPS ON HOW TO MAKE A MODEL OF THE SOLAR SYSTEM

i. Model planets and the sun – You can use clay, paper mache, plasticine or candle wax.
ii. Paste Manila paper on a soft board.
iii. Draw circles on the Manila paper to show the orbits.
iv. Use pins or thorns to mount the planets on the orbits and the sun at the centre.
v. Put the name tags against the sun and the planets.

N.B

- Stars that are very far and cannot be seen with the naked eye can be observed using a Telescope.
- Distance between the earth and the sun is 150 million kilometers.
- Meteors - These are rocks orbiting in space within the solar system and they blaze (burn) as they move across the sky.
  - Meteors rub against the air causing friction and making them burn/blaze.
  - A meteor that is seen to blaze/burn across the sky as a result of the friction of air rubbing against it is called a shooting star.
- Meteorites - meteors that reach the earth's surface. Sometimes they cause big holes/craters.
- Artificial satellites - bright star-like artificial objects placed in the sky to move round the earth, with a specific purpose e.g. communication, forecasting, scouting, etc.